DSC 291: Learning Theory - PA3

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Part A: Mini Course on Concentration Inequalities

1. The concept of concentration.

Concentration inequalities control fluctuations of a random quantity around its mean (or median).
Typical form: if Z depends on many random inputs, then

$$\Pr\left(|Z - \mathbb{E}Z| \geq t\right) \leq \mathrm{small}(t).$$

The phrase *concentrates* means this tail probability decays quickly (often exponentially in t^2 or t).

For learning, the central object is the empirical risk

$$L_n(h) = \frac{1}{n}\sum_{i=1}^{n} \ell(h; Z_i), \qquad Z_i \overset{i.i.d.}{\sim} \mathcal{D},\ \ell \in [0,1].$$

Since $L_n(h)$ is an average of bounded random variables, we expect it to be close to

$$L_{\mathcal{D}}(h) = \mathbb{E}[\ell(h; Z)]$$

with high probability. Concentration inequalities make this precise.

Concrete example (coin flips). If $X_i \in \{0,1\}$ are Bernoulli(p), then $\bar{X}_n = \frac{1}{n}\sum X_i$ estimates p. Hoeffding[1] gives

$$\Pr\left(|\bar{X}_n - p| \geq \varepsilon\right) \leq 2e^{-2n\varepsilon^2}.$$

So error ε has probability roughly $e^{-n\varepsilon^2}$: doubling sample size dramatically shrinks tail risk.

Intuition. Independent noise terms cancel. To make an average deviate by t, many terms must align in the same unfavorable direction, an exponentially unlikely event.

2. The common technique.

The unifying mechanism is the **MGF/Chernoff method**.

Let Z be any random variable and $\lambda > 0$. By Markov,

$$\Pr(Z \geq t) = \Pr(e^{\lambda Z} \geq e^{\lambda t}) \leq e^{-\lambda t}\, \mathbb{E}[e^{\lambda Z}].$$

Hence

$$\Pr(Z \geq t) \leq \inf_{\lambda > 0} \exp\big(-\lambda t + \log \mathbb{E} e^{\lambda Z} \big).$$

So a tail bound is obtained once we control the log-MGF $\psi_Z(\lambda) := \log \mathbb{E} e^{\lambda Z}$.

For sums $Z = \sum_i Y_i$ of independent variables, MGFs factor:

$$\psi_Z(\lambda) = \sum_i \log \mathbb{E}[e^{\lambda Y_i}],$$

which reduces the proof to one-variable bounds.

- Hoeffding: uses boundedness $Y_i \in [a_i, b_i]$ to show sub-Gaussian log-MGF.
- Bernstein: uses boundedness **and** variance information to get a refined log-MGF and sharper small-deviation behavior.

For functions $f(X_1, \ldots, X_n)$ (not necessarily sums), the martingale variant applies:

1. Define Doob martingale $M_i = \mathbb{E}[f \mid X_1, \ldots, X_i]$.
2. Bound increments $|M_i - M_{i-1}|$ by bounded differences constants.
3. Apply exponential supermartingale/Chernoff to $M_n - M_0$.

This yields McDiarmid's inequality.

3. Statements, assumptions, and consequences.

Let $X_1, \ldots, X_n$ be random variables and $S_n = \sum_{i=1}^n (X_i - \mathbb{E}X_i)$.

| Inequality | Assumptions | Controls | Typical tail | |---|---|---|---| | Hoeffding | independent, $X_i \in [a_i, b_i]$ a.s. | sum/mean of bounded independent RVs | $\Pr(S_n \geq t) \leq \exp\left(-\frac{2t^2}{\sum_i (b_i - a_i)^2}\right)$ | | Hoeffding (without replacement) | sample uniformly without replacement from finite population in $[a, b]$ | sample mean from finite population | at least as concentrated as i.i.d. sampling; e.g. $\Pr(\bar{X} - \mu \geq \varepsilon) \leq e^{-2n\varepsilon^2/(b-a)^2}$ (and sharper Serfling finite-population correction) | | McDiarmid | independent inputs, bounded differences: changing coordinate i changes f by at most c_i | general function $f(X_1, \ldots, X_n)$ | $\Pr(f - \mathbb{E}f \geq t) \leq \exp\left(-\frac{2t^2}{\sum_i c_i^2}\right)$ | | Bernstein | independent, centered, $|X_i| \leq b$, variance proxy $V = \sum \mathbb{E}[X_i^2]$ | sum of bounded RVs with variance info | $\Pr(S_n \geq t) \leq \exp\left(-\frac{t^2}{2(V + bt/3)}\right)$ |

Comparison.

- Hoeffding uses only support bounds, so it is distribution-agnostic but can be conservative.
- Bernstein adds variance: when variance is small, it improves the rate (sub-Gaussian for moderate t, sub-exponential only in the far tail).
- McDiarmid extends concentration from sums to *stable functions* of independent inputs.
- Sampling without replacement gains negative dependence; concentration is no worse (often better) than i.i.d. sampling.

4. Proofs.

4.1 Hoeffding (independent bounded case)

If $Y \in [m, M]$ and $\mathbb{E}Y = 0$, Hoeffding's lemma gives

$$\log \mathbb{E}e^{\lambda Y} \leq \frac{\lambda^2 (M-m)^2}{8}.$$

For $Y_i = X_i - \mathbb{E}X_i$ (independent),

$$\log \mathbb{E}e^{\lambda S_n} \leq \sum_i \frac{\lambda^2 (b_i - a_i)^2}{8}.$$

Chernoff then gives

$$\Pr(S_n \geq t) \leq \exp\left(-\lambda t + \frac{\lambda^2}{8} \sum_i (b_i - a_i)^2\right).$$

Optimizing in λ yields

$$\Pr(S_n \geq t) \leq \exp\left(-\frac{2t^2}{\sum_i (b_i - a_i)^2}\right). \ \square$$

4.2 Hoeffding for sampling without replacement

Let $(X_1, \ldots, X_n)$ be sampled without replacement from a finite multiset $\{v_1, \ldots, v_N\} \subset [a, b]$, and let $(Y_1, \ldots, Y_n)$ be i.i.d. draws with replacement from the same population. Hoeffding (1963) proves convex domination

$$\mathbb{E}\,\phi\left(\sum_{i=1}^n X_i\right) \leq \mathbb{E}\,\phi\left(\sum_{i=1}^n Y_i\right)$$

for every convex ϕ. Taking $\phi(u) = e^{\lambda u}$ gives MGF domination; therefore every Chernoff tail for i.i.d. bounded sampling also holds without replacement. Applying Hoeffding's i.i.d. bound to Y_i yields

$$\Pr(\bar{X} - \mu \geq \varepsilon) \leq \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right). \ \square$$

(Serfling's refinement multiplies exponent by a finite-population correction factor.)

4.3 McDiarmid (bounded differences)

Assume independent $X_1, \ldots, X_n$ and

$$\sup_{x_1,\ldots,x_n,x_i'} |f(x_1,\ldots,x_i,\ldots,x_n) - f(x_1,\ldots,x_i',\ldots,x_n)| \leq c_i.$$

Define $M_i = \mathbb{E}[f(X_1,\ldots,X_n) \mid X_1,\ldots,X_i]$. Then (M_i) is a martingale and $|M_i - M_{i-1}| \leq c_i$. Applying Hoeffding's lemma conditionally to each increment gives

$$\log \mathbb{E}e^{\lambda(M_n - M_0)} \leq \frac{\lambda^2}{8} \sum_i c_i^2.$$

Chernoff optimization yields

$$\Pr\left(f - \mathbb{E}f \geq t\right) \leq \exp\left(-\frac{2t^2}{\sum_i c_i^2}\right). \ \square$$

4.4 Bernstein

Let independent centered X_i satisfy $|X_i| \leq b$ a.s., and set $V = \sum_i \mathbb{E}[X_i^2]$. A standard mgf bound is

$$\log \mathbb{E}e^{\lambda X_i} \leq \frac{\lambda^2 \mathbb{E}[X_i^2]}{2(1 - \lambda b/3)}, \qquad 0 < \lambda < 3/b.$$

Summing over i:

$$\log \mathbb{E}e^{\lambda S_n} \leq \frac{\lambda^2 V}{2(1 - \lambda b/3)}.$$

Then

$$\Pr(S_n \geq t) \leq \inf_{0<\lambda<3/b} \exp\left(-\lambda t + \frac{\lambda^2 V}{2(1 - \lambda b/3)}\right)$$

and optimization gives

$$\Pr(S_n \geq t) \leq \exp\left(-\frac{t^2}{2(V + bt/3)}\right). \ \square$$

Common template across all four proofs: exponential tilt + MGF control + optimization in λ. The only difference is *how* the MGF is controlled (bounded range, variance refinement, martingale increments, or negative dependence).

5. Connection to the new ERM proof.

We establish the connection to the new ERM proof via one of the following methods *(Note: Method 1 is more traditional, Method 2 may be more intuitive depending on the reader)*:

Method 1: Using Hoeffding's Inequality and Bernstein's Inequality

Let losses be in $[0,1]$. Define

$$h^* \in \arg\min_{h\in\mathcal{H}} L_{\mathcal{D}}(h), \qquad g_h(z) := \ell(h;z) - \ell(h^*;z) \in [-1,1].$$

Then $\Delta(h) := \mathbb{E}[g_h] = L_{\mathcal{D}}(h) - L_{\mathcal{D}}(h^*) \geq 0$, and for ERM $\hat{h} \in \arg\min_h L_n(h)$ we have

$$\Delta(\hat{h}) = L_{\mathcal{D}}(\hat{h}) - L_{\mathcal{D}}(h^*).$$

For 0-1 loss (and similarly for bounded losses with a constant factor), the excess variable satisfies a Bernstein condition

$$\mathrm{Var}(g_h) \leq \mathbb{E}[g_h] = \Delta(h).$$

Applying Bernstein to fixed h and then a two-sample symmetrization/union bound over labelings on $2n$ points gives, with probability at least $1-\delta$, simultaneously for all $h \in \mathcal{H}$,

$$\Delta(h) \leq 2\big(L_n(h) - L_n(h^*)\big) + C\frac{\log \Gamma_{\mathcal{H}}(2n) + \log(1/\delta)}{n}$$

for a universal constant $C > 0$.

Now plug in $h = \hat{h}$. Since $\hat{h}$ minimizes L_n,

$$L_n(\hat{h}) - L_n(h^*) \leq 0.$$

Therefore

$$L_{\mathcal{D}}(\hat{h}) - L_{\mathcal{D}}(h^*) = \Delta(\hat{h}) \leq C\frac{\log \Gamma_{\mathcal{H}}(2n) + \log(1/\delta)}{n}.$$

Equivalently,

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_{h\in\mathcal{H}} L_{\mathcal{D}}(h) + O\left(\frac{\log \Gamma_{\mathcal{H}}(2n) + \log(1/\delta)}{n}\right).$$

This is the announced i.i.d. growth-function ERM guarantee.

Method 2: Using Ghost Samples

Let losses be in $[0,1]$. Define

$$h^* \in \arg\min_{h\in\mathcal{H}} L_{\mathcal{D}}(h), \qquad g_h(z) := \ell(h;z) - \ell(h^*;z) \in [-1,1].$$

Then $\Delta(h) := \mathbb{E}[g_h] = L_{\mathcal{D}}(h) - L_{\mathcal{D}}(h^*) \geq 0$, and for ERM $\hat{h} \in \arg\min_h L_n(h)$ we have

$$\Delta(\hat{h}) = L_{\mathcal{D}}(\hat{h}) - L_{\mathcal{D}}(h^*).$$

To control $\Delta(h)$ uniformly we follow the standard symmetrization + counting pipeline.

Symmetrization. [7] Draw an independent ghost sample $S' = (Z_{n+1}, \ldots, Z_{2n})$ from $\mathcal{D}^n$ and let $L'_n(h)$ be the empirical risk on S'. For every class of $[0,1]$ losses (hence for g_h) we have

$$\Pr\Big(\sup_{h\in\mathcal{H}}\big(L_{\mathcal{D}}(h) - L_n(h)\big) > \varepsilon\Big) \le 2\,\Pr\Big(\sup_{h\in\mathcal{H}}\big(L'_n(h) - L_n(h)\big) > \varepsilon/2\Big).$$

Condition on (S, S'). For fixed h, $L'_n(h) - L_n(h)$ is an average of $2n$ centered, bounded variables, so Bernstein (using $\mathrm{Var}(g_h) \le \Delta(h) \le 1$) gives

$$\Pr\Big(L'_n(h) - L_n(h) > t \mid S, S'\Big) \le \exp\big(-cnt^2\big)$$

for some universal $c > 0$.

Counting via $\Gamma_{\mathcal{H}}(2n)$. The gap $L'_n(h) - L_n(h)$ depends only on the restriction of h to the $2n$ points in $S \cup S'$. Hence, conditioned on (S, S'), there are at most $\Gamma_{\mathcal{H}}(2n)$ distinct values of the gap. A union bound gives

$$\Pr\Big(\sup_{h\in\mathcal{H}}|L'_n(h) - L_n(h)| > t \mid S, S'\Big) \le \Gamma_{\mathcal{H}}(2n)\exp\big(-cnt^2\big)\,.$$

Integrating over (S, S') and combining with symmetrization yields (with probability $\ge 1-\delta$) the uniform bound

$$\sup_{h\in\mathcal{H}}|L_{\mathcal{D}}(h) - L_n(h)| \le \sqrt{\frac{1}{cn}\Big(\log\Gamma_{\mathcal{H}}(2n) + \log\frac{2}{\delta}\Big)}.$$

Plugging in ERM optimality. On the above $1-\delta$ event,

$$L_{\mathcal{D}}(\hat{h}) \le L_n(\hat{h}) + \sqrt{\frac{1}{cn}\Big(\log\Gamma_{\mathcal{H}}(2n) + \log\frac{2}{\delta}\Big)} \le L_n(h^*) + \sqrt{\frac{1}{cn}\Big(\log\Gamma_{\mathcal{H}}(2n) + \log\frac{2}{\delta}\Big)}.$$

A second application of the uniform deviation bound to h^* replaces $L_n(h^*)$ by $L_{\mathcal{D}}(h^*)$, resulting in

$$L_{\mathcal{D}}(\hat{h}) \le L_{\mathcal{D}}(h^*) + 2\sqrt{\frac{1}{cn}\Big(\log\Gamma_{\mathcal{H}}(2n) + \log\frac{2}{\delta}\Big)}.$$

Absorbing constants into $O(\cdot)$ gives precisely the stated growth-function ERM guarantee.

Part B

1. Proof.

We prove the updated No-Free-Lunch theorem.

Quantifiers. The learner A and integer $n < |\mathcal{X}|/2$ are fixed universally. The adversary then chooses a hard distribution $\mathcal{D}$.

Choose distinct points $C = \{x_1, \ldots, x_{2n}\} \subset \mathcal{X}$. For each labeling $\sigma \in \{0,1\}^{2n}$ define $f_\sigma(x_i) = \sigma_i$. Let $\mathcal{D}_\sigma$ be uniform on

$$\{(x_i, f_\sigma(x_i)) : 1 \le i \le 2n\}.$$

Clearly $L_{\mathcal{D}_\sigma}(f_\sigma) = 0$ (realizable).

Let $S \sim \mathcal{D}_\sigma^n$. For fixed sample points, define unseen set

$$U(S) := \{i \in [2n] : x_i \text{ does not appear in } S\}.$$

Conditioned on sample points, labels on $U(S)$ are independent fair bits under uniform random σ, while $A(S)$ is independent of those unseen bits. Hence each unseen point contributes error probability $1/2$. Therefore

$$\mathbb{E}_\sigma\big[L_{\mathcal{D}_\sigma}(A(S)) \mid S\big] = \frac{|U(S)|}{4n}.$$

Taking expectation over S:

$$\mathbb{E}_{\sigma,S}\, L_{\mathcal{D}_\sigma}(A(S)) = \frac{\mathbb{E}|U(S)|}{4n}.$$

Now

$$\mathbb{E}|U(S)| = \sum_{i=1}^{2n} \Pr(x_i \text{ unseen}) = 2n\left(1 - \frac{1}{2n}\right)^n \ge 2ne^{-1/2}.$$

So

$$\mu := \mathbb{E}_{\sigma,S}\, L_{\mathcal{D}_\sigma}(A(S)) \ge \frac{e^{-1/2}}{2} > 0.3.$$

By averaging, there exists at least one labeling σ^* such that

$$\mathbb{E}_S\, L_{\mathcal{D}_{\sigma^*}}(A(S)) \ge \mu.$$

Set $Y := L_{\mathcal{D}_{\sigma^*}}(A(S)) \in [0,1]$. Using

$$\mathbb{E}[Y] \le a\, \Pr(Y < a) + 1 \cdot \Pr(Y \ge a) = a + (1-a)\Pr(Y \ge a),$$

we get

$$\Pr(Y \ge a) \ge \frac{\mathbb{E}[Y] - a}{1 - a}.$$

With $a = 1/8$ and $\mathbb{E}[Y] \ge 0.3$,

$$\Pr\left(Y \geq \frac{1}{8}\right) \geq \frac{0.3 - 0.125}{0.875} = 0.2 \geq \frac{1}{7}.$$

Thus for $\mathcal{D} = \mathcal{D}_{\sigma^*}$: (i) realizable, and (ii) with probability at least $1/7$, $L_{\mathcal{D}}(A(S)) \geq 1/8$. $\square$

2. Application: closing the Fundemental Theorem.

Corollary (NFL $\Rightarrow$ lower-bound direction). If $\mathrm{VCdim}(\mathcal{H}) = \infty$, then $\mathcal{H}$ is not PAC learnable (distribution-free, realizable binary classification).

Reason. Fix any learner A and sample size n. Since $\mathrm{VCdim}(\mathcal{H}) = \infty$, pick a shattered set of size $2n$. Restricting $\mathcal{H}$ to that set realizes all 2^{2n} labelings, so the NFL theorem applies and gives a realizable distribution with

$$\Pr\left(L_{\mathcal{D}}(A(S)) \geq 1/8\right) \geq 1/7.$$

Hence no finite n can guarantee arbitrary (ε, δ) (e.g., $\varepsilon < 1/8$, $\delta < 1/7$), so PAC learnability fails.

Concrete infinite-VC class. Let $\mathcal{X} = \mathbb{N}$ and

$$\mathcal{H} = \{0,1\}^{\mathbb{N}} \quad (\text{all binary functions on } \mathbb{N}).$$

For any finite set $S = \{x_1, \ldots, x_m\} \subset \mathbb{N}$ and any labels $y \in \{0,1\}^m$, define $h_y \in \mathcal{H}$ by $h_y(x_i) = y_i$ and arbitrary values elsewhere. So every finite S is shattered, therefore $\mathrm{VCdim}(\mathcal{H}) = \infty$. Applying the corollary gives that this class is not PAC learnable.

3. Worked construction.

Take $n = 1$, domain $\mathcal{X} = \{a, b, c\}$ (so $n < |\mathcal{X}|/2$), and learning rule:

- On sample $((x, y))$, output hypothesis $h_{x,y}$ with $h_{x,y}(x) = y$ and $h_{x,y}(x') = 0$ for $x' \neq x$.

Choose target $f^*(a) = f^*(b) = 1$ (and any value on c), and distribution $\mathcal{D}$ uniform on $\{(a,1), (b,1)\}$. Then $L_{\mathcal{D}}(f^*) = 0$ (realizable).

If sample point is a, learner predicts $h(a) = 1, h(b) = 0$; if sample point is b, learner predicts $h(a) = 0, h(b) = 1$. In either case, exactly one of a, b is misclassified, each with mass $1/2$. Therefore

$$L_{\mathcal{D}}(A(S)) = \frac{1}{2} \quad \text{for every sample } S.$$

Hence

$$\Pr_{S \sim \mathcal{D}}\left(L_{\mathcal{D}}(A(S)) \geq \frac{1}{8}\right) = 1.$$

This explicitly exhibits a bad distribution/target pair for this learner, with constant (indeed unit) failure probability.

4. Comparison to the original NFL theorem.

Aspect	Original theorem (mistake bound)	Updated theorem (PAC/ statistical)
Setting	Online/sequential prediction on a finite domain	Batch statistical learning under i.i.d. sampling
Adversary	Chooses an ordering/sequence and labels to force mistakes (can be adaptive in online framing)	Chooses a fixed distribution $\mathcal{D}$ (oblivious) before sampling
Learner type	Deterministic learner in sequence model	Any learner (possibly randomized) in PAC model
Conclusion type	Deterministic worst-case number of mistakes on a sequence	Probabilistic lower bound on population loss over random sample
Quantity lower-bounded	Total mistakes (discrete count)	Risk $L_{\mathcal{D}}(A(S))$ (expected 0-1 error under data distribution)

Interpretation.

- Original NFL: without structural assumptions, there exists a sequence where the learner performs catastrophically in online prediction.
- Updated NFL: without finite-capacity assumptions (finite VC dimension), there exists a realizable distribution on which learning cannot uniformly guarantee small generalization error.

Both are called "No-Free-Lunch" because each says: absent inductive bias/complexity control, there is no universally successful learner.

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Part C: AI Usage Report

1. Due to this week's changes to the way Claude works, as well as Anthropic's pricing tiers, Codex 5.1 was used to explore theorems and lemmas. `AGENTS.md` and `PLAN.md` was used to create the proof sketch, along with these workflow addendums:

    - Normally, Codex 5.3 would be used. Even though GPT 5.4 has been out since March 5, 2026, there is no Codex line of models yet, OpenAI claims to have integrated Codex 5.3 into GPT 5.4. Still, consumer consensus on Hacker News show that OpenAI may be deliberatly throttling the performance of these models to save on compute costs, and to improve on training the newer models by making 5.4 stupid on purpose. Codex 5.1 still has massive parameters, so it is suitable for this proof sketch.
    - Claude's performace across their Sonnet and Opus models have noticably been getting worse. It turns out that Anthropic was indeed degrading their reasoning performance on their models and have since implemented hotfixes. In the meantime, the API pricing still remains too high after Anthropic's promotion ended. TLDR: Claude sucks this week, its degraded reasoning means that its proof construction abilities will suffer immensely on topics like PAC and ERM.
    - Roo Code has custom modes. In the past, the role of the agent was included directly in `AGENTS.md` , but the role of 'Professional' could potentially conflict with 'Architect', 'Code', and 'Ask'. Thus, 'mathematics-professor' was made where the `persona` , `capabilities` , and `operational rules` could be affixed under `Role Definition` . This helps in minimizing tokens to reduce the use of the extended context size alloted to Codex, with the reasoning being that shifted sparse attention effectively enables context extension [8] which is great for proof writing. The 'mathematics-professor' came up with reasonable proofs and self checked its work before writing to `PLAN.md`
    - The other `Role Definition` generated was 'phd-student-researcher' to verify the work of the 'mathematics-professor' for logical gaps that may not follow for a beginner. The student was able to suggest alternative lemmas, some of which were

hallucinated and thus required manual human review. But overall, the 'student' agreed with the 'professor' for the most part. It is unclear if the agreement is due to the proofs being logically structured or if it is because Codex 5.1 was used for both agents that resided in the same context window, but I am inclined to believe it is the former since I am able to follow the proof.
    - `AGENTS.md` has been modified to better assist in the creation of `PLAN.md`, allowing for more detailed reports and less mistakes. Different roles can check each other's work using `PLAN.md` as it provides better context to `AGENTS.md` and the output presented in this notebook.
    - Codex 5.1 API pricing: $3.50
2. The comparisons of the different No Free Lunch Theorems were accepted, as it showed a high level overview of what each version of NFL is able to prove. Specifically, it showed that the updated NFL was able to prove how ERM is PAC learnable. [9, 10]

3. Codex 5.1 has hallcuinated lemmas. Specifically, it claimed that Lemma 2.3 from R. Boucheron et al. (2012) could support the ERM guarantee in Part A. A review of R. Boucheron et al. (2012) proved Lemma 2.3 does not exist; there was however a Lemma 2.2 (Hoeffding's lemma) and Lemma 2.4 ([11]). Therefore this proof sketch for the ERM guarantee was rejected in favor of using ghost samples as described in F. Lauer (2014) and the correct Hoeffding's Inequality Theorem. Two separate agents claimed that while the ghost sample apprach was correct, Hoeffding and Bernstein were better suited; regardless both proof sketches were kept for reference. Citations underwent manual review to verify their existence and comply with IEEE standards.

4. The inital proof sketches were drafted with a "mathematics-professor" agent, and were checked with a "phd-student-researcher" agent. We also manually reviewed the proof sketches ourselves for any logic jumps that needed to be explained more in-depth.